

DVN-SLAM: Dynamic Visual Neural SLAM Based on Local-Global Encoding

Wenhua Wu¹, Guangming Wang², Ting Deng³, Sebastian Aegidius⁴, Stuart Shanks⁴,
Valerio Modugno⁴, Dimitrios Kanoulas⁴, Hesheng Wang^{1*}

Abstract—Recent research on Simultaneous Localization and Mapping (SLAM) based on implicit representation has shown promising results in indoor environments. However, some challenges remain: the limited scene representation capability of implicit encoding, the uncertainty in the rendering process from implicit representations, and the disruption of consistency by dynamic objects. To address these challenges, we propose a dynamic visual SLAM system based on local-global fusion neural implicit representation, named DVN-SLAM. To improve the scene representation capability, we introduce a local-global fusion neural implicit representation that enables the construction of an implicit map while considering both global structure and local details. To tackle uncertainties arising from the rendering process, we design an information concentration loss for optimization, aiming to concentrate scene information on object surfaces. The proposed DVN-SLAM achieves competitive performance in localization and mapping across multiple datasets. More importantly, DVN-SLAM demonstrates robustness without semantic and optical flow prior in dynamic scenes, which sets it apart from other NeRF-based methods.

I. INTRODUCTION

Dense visual Simultaneous Localization and Mapping (SLAM) is the foundation of perception, navigation, and planning that finds wide applications in areas such as autonomous driving, mobile robotics, and virtual reality. Traditional visual SLAM methods primarily focus on localization accuracy and utilize explicit geometric primitives such as surfaces, voxels, and point clouds to represent environmental information [1], [2]. However, these explicit representations often suffer from noise and holes. In recent years, the rapid development of Neural Radiance Field (NeRF) has demonstrated its tremendous potential in scene modeling [3], [4], [5], [6], [7], [8]. NeRF employs implicit encoding and Multi-Layer Perceptron (MLP) to map spatial information, enabling dense and flexible scene construction.

Despite the impressive performance of NeRF-based SLAM in static indoor scenes, there are still several open challenges. Firstly, the scene modeling capability of existing methods is limited. iMAP [9] utilizes a position-encoded neural implicit representation that suffers from overly smooth local details and forgetfulness as the scene size increases.

*This work was supported in part by the Natural Science Foundation of China under Grant 62225309, U24A20278, 62361166632, U21A20480, and the UKRI FLF [MR/V025333/1] (RoboHike). Corresponding Author: Hesheng Wang, wanghesheng@sjtu.edu.cn.

¹Shanghai Jiao Tong University

²University of Cambridge

³Imperial College London

⁴University College London

Methods based on feature grids or planes [10], [11] provide precise modeling of local scene details but exhibit a significant decrease in global representation and predictive ability. Co-SLAM [12] combines sparse hash encoding and coordinate encoding, but the joint is simplistic and does not fully exploit their respective advantages.

Furthermore, existing NeRF-based SLAM methods struggle with dynamic scenes. The motion of objects disrupts the static consistency of the scene, making the pure pose implicit mapping inadequate for modeling. Although some methods attempt to filter out dynamic objects, they all rely on pre-trained semantic or optical flow networks [13], [14], [15], [16], [17]. This requires a significant amount of computational resources and poses a risk of false detection. In this paper, we present a novel local-global fusion neural implicit representation that effectively leverages the advantages of the continuous neural radiance field and the discrete grid, enabling the modeling of both local details and global structure. More importantly, the designed neural implicit representation can maintain stable scene structure modeling in dynamic scenarios. Rapidly moving objects can be automatically ignored without semantic and optical flow prior, and the background occluded by dynamic objects can be effectively recovered.

The optimization of neural radiance fields relies on supervised losses between rendered images and observed images [3]. The process of volume rendering introduces certain uncertainties, where different information distributions along the viewing ray can yield the same rendering result upon integration. This uncertainty implies that even with minimal rendering errors, the distribution of scene information may not be accurate. To address this, we design an information concentration loss based on rendering variance, aiming to concentrate information along the observed rays. This aligns with the prior knowledge that color information tends to be concentrated on surfaces.

The main contributions can be summarized as follows:

- We propose DVN-SLAM, a dynamic robust dense visual SLAM system. The core is a local-global fusion neural implicit representation, which employs attention-based feature fusion and result fusion to leverage the advantages of the local discrete grid and the global continuous neural radiance field.
- We consider the uncertainty of rendering and design an information concentration loss for optimization. This makes scene modeling more accurate and inhibits unnecessary reconstruction.

- Extensive experimental results demonstrate that DVN-SLAM not only achieves competitive performance in static scenes, but also remains effective in high-dynamic scenes without semantic and optical flow prior.

II. RELATED WORK

A. Dense Visual SLAM Prior to NeRF

Visual SLAM systems focus on localizing and mapping by collecting data with a monocular camera. The foundation of dense visual SLAM lies in DTAM [18], which pioneers real-time tracking and mapping via dense scene representation. In the same period, the KinectFusion method [19] makes notable strides by using ICP algorithms [20] and volumetric Truncated Signed Distance Function (TSDF) to achieve accurate and real-time reconstruction of dense surfaces for indoor scenes. Lots of innovative data structures including Surfels [21], [22] and Octrees [23], [24], are proposed to improve scalability and reduce memory. In contrast to these methods which rely on per-frame pose optimization, BAD-SLAM [25] is the first to propose a full Bundle Adjustment (BA) to jointly optimize the keyframes. Recently, numerous deep learning-based methods [26], [27], [28], [29], [30], [31], [32] are introduced to improve the precision and robustness with various learnable parameters.

B. Neural Implicit Representations

Neural Radiance Field (NeRF) [3] utilizes an MLP to model the mapping from spatial locations to the geometry and appearance information, achieving remarkable results through volume rendering. The initial NeRF [3] could only model static individual objects, while subsequent methods have studied the implicit representation of challenging scenes involving large scale [33], [34], [35], [36], [37], dynamics [5], [38], [39], lighting variations [6], [40], and other factors. Furthermore, some methods investigate how to enhance the learning efficiency of implicit representation [41].

C. Neural Implicit SLAM

Recent work on 3D scene reconstruction implies the advantage of both SLAM and NeRF, achieving an accurate and real-time reconstruction for static indoor scenes [42]. One classical work is iMAP [9], which performs jointly real-time mapping and tracking, plausible filling to the unobserved regions via trained MLP. Subsequently, NICE-SLAM [10] improves the iMAP [9] model by introducing a hierarchical feature grid for scene representation and optimizing it using pre-trained decoders. PLGSLAM [43], Point-SLAM [44], Co-SLAM [12] and ESLAM [11] propose various approaches to implicit scene representation. Some methods [45], [46], [47] are based on 3D Gaussians splatting [48], which makes scene rendering more efficient. However, the above methods can only work in static scenes. Existing dynamic implicit slam methods all rely on pre-trained semantic or optical flow networks to filter out dynamic objects [13], [14], [15], [16]. This requires a significant amount of computational resources and poses a risk of false detection.

III. METHOD

The overview of the proposed DVN-SLAM is shown in Fig. 1. Given the RGB-D stream, our method predicts the camera poses and builds an implicit map of the scene. Sec. III-A introduces the local-global fusion neural implicit representation. Sec. III-B shows loss functions. Sec. III-C describes the tracking and mapping optimization process.

A. Global-Local Fusion Neural Implicit Representation

The proposed local-global fusion neural implicit representation employs both attention-based feature fusion and result fusion, aiming to leverage the advantages of continuous neural radiance fields and discrete feature planes while minimizing their drawbacks. In detail, we employ the One-blob [49] encoding $f_o(p)$ for global representation. For local representation, we use hierarchical axis-aligned feature planes, with coarse and fine geometric feature planes $f_g^l = \{F_{g-XY}^l, F_{g-XZ}^l, F_{g-YZ}^l\}$, and color feature planes $f_a^l = \{F_{a-XY}^l, F_{a-XZ}^l, F_{a-YZ}^l\}$. Given the point coordinate $p = (x, y, z)$, we query their components on each plane and sum them to obtain local geometric feature $f_g(p)$ and appearance feature $f_a(p)$:

$$\begin{aligned} f_i^l(p) &= F_{i-XY}^l(x, y) + F_{i-XZ}^l(x, z) + F_{i-YZ}^l(y, z), \\ f_i(p) &= [f_i^c(p), f_i^f(p)], \end{aligned} \quad (1)$$

where, the superscripts $l \in \{c, f\}$ mean coarse and fine levels. The subscripts $i \in \{a, g\}$ mean appearance and geometric. $\{XY, XZ, YZ\}$ are coordinate planes. The operation of $[\cdot, \cdot]$ denotes feature concatenation.

Attention-based feature fusion. We concatenate the global features and local features and use an attention module for fusion:

$$\begin{aligned} f_i^{cat}(p) &= [f_i(p), f_o(p)], \\ f_i^{att}(p) &= \text{Softmax}\left(\frac{f_i^{cat}(p) f_i^{cat}(p)^T}{\sqrt{d_i}}\right) f_i^{cat}(p), \end{aligned} \quad (2)$$

where, d_i is the dimension of $f_i^{cat}(p)$. $f_i^{att}(p)$ is the final fused feature. The attention fusion combines different features of the same point, including the grid feature from local representation and the pose encoding feature from global representation, enabling better fusion of local and global representations than direct concatenation.

Result fusion. We use MLPs to decode the fusion features and local features respectively, and then fuse the results to get the final geometry and color information:

$$c_p = \omega \phi_a^1(f_a^{att}(p)) + (1 - \omega) \phi_a^2(f_a(p)), \quad (3)$$

$$s_p = \omega \phi_g^1(f_g^{att}(p)) + (1 - \omega) \phi_g^2(f_g(p)), \quad (4)$$

where $\{\phi_a^1, \phi_a^2, \phi_g^1, \phi_g^2\}$ are feature decoders. ω is the fusion weight. c_p is the RGB value, and s_p is the TSDF value.

It is important to note that although Co-SLAM [12] also uses both coordinate and hash encodings, it simply concatenates them together. In contrast, our approach combines discrete local features and continuous coordinate encoding through attention-based feature fusion and result fusion, achieving a better balance between local scene details and

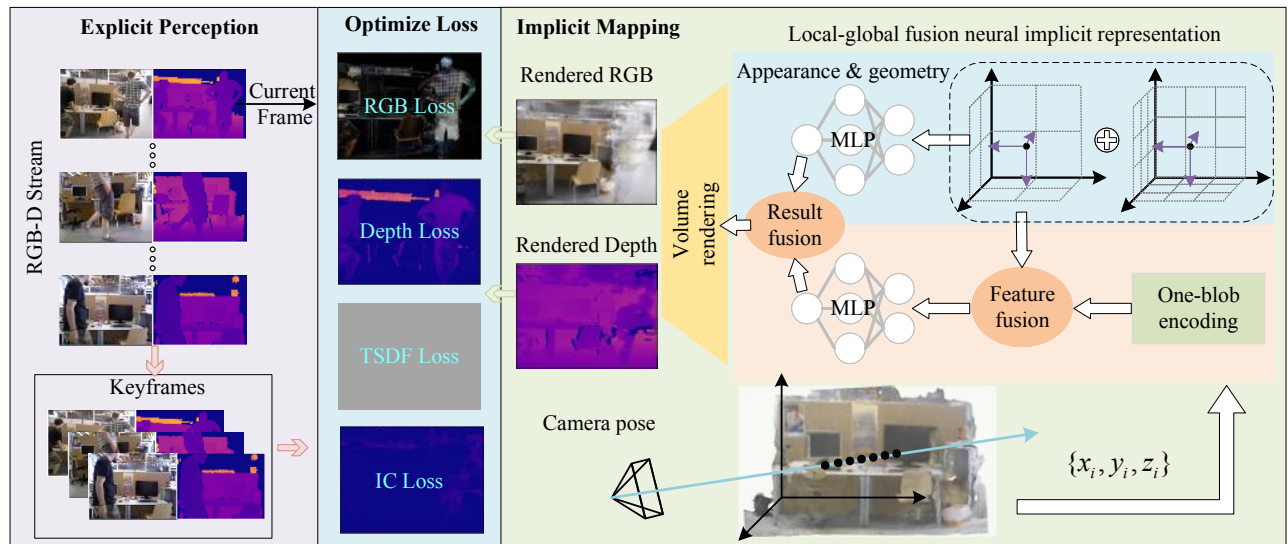


Fig. 1. Overview of DVN-SLAM. The left side of the framework is the explicit perception module, which takes RGB-D video as input. The keyframe selection is based on the information gain from the image. On the right side of the framework is the local-global fusion neural implicit representation, which establishes a mapping from spatial locations to color and TSDF values. Volume rendering is performed to generate RGB-D images corresponding to the tracking poses. The middle is our loss optimization module, which includes color loss, depth loss, TSDF loss, and information concentration loss.

hole-filling. More importantly, this design makes our method insensitive to dynamic objects and able to maintain the scene structure modeling while other methods are ineffective.

Color and depth rendering. We follow the same sampling strategy as ESLAM [11], where N points are sampled along each ray. The implicit representation model predicts the color $\{c_i\}$ and TSDF $\{s_i\}$ for each sampled point. We employ the following transformation function to convert the TSDF into weights $\{w_i\}$:

$$w_i = \sigma\left(\frac{s_i}{tr}\right)\sigma\left(-\frac{s_i}{tr}\right), \quad (5)$$

where tr is the truncation distance and σ represents the Sigmoid function. This direct weight computation ensures that points on the same ray do not interfere with each other. When the TSDF values of some points become abnormal due to dynamic disturbances, the weights of other points are not affected. In contrast, methods that calculate weights by integrating along the ray would lead to abnormal weights for all points following the occurrence of abnormal points.

Then, the color value and depth value of the pixel are obtained by weighted summing the color and depth of the sampled points on the ray:

$$\hat{c} = \frac{1}{\sum_{i=1}^N w_i} \sum_{i=1}^N w_i c_i, \hat{d} = \frac{1}{\sum_{i=1}^N w_i} \sum_{i=1}^N w_i d_i. \quad (6)$$

B. Loss Functions

Color and depth loss. In order to reduce the computational burden, we randomly sample M points for each image to render, and then calculate the supervision loss of rendering

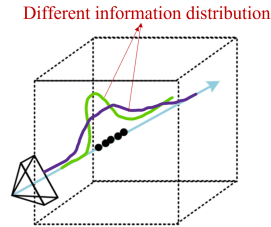


Fig. 2. Illustration of Volume Rendering. Different information distributions along the same ray may yield the same rendering result.

and observing RGB-D images:

$$L_{rgb} = \frac{1}{M} \sum_{m=1}^M (\hat{c}_m - c_m)^2, \quad (7)$$

$$L_d = \frac{1}{M_d} \sum_{m=1}^{M_d} (\hat{d}_m - d_m)^2,$$

where M_d is the number of pixels where the depth value is valid. $M_d \leq M$, because there are holes in the depth map.

Information concentration loss. As shown in Fig. 2, the volumetric rendering process is inherently uncertain, as there is a possibility that different information distributions along a ray can yield the same rendered result. Due to this uncertainty, even with small rendering loss, there is no guarantee that the learned scene information is accurate. To address this, we propose a depth-variance-based information concentration loss:

$$L_{ic} = \frac{1}{M_d} \sum_{m=1}^{M_d} \frac{1}{\sum_{i=1}^N w_i} \sum_{i=1}^N w_i (d_i - \hat{d})^2. \quad (8)$$

By optimizing this loss, we encourage the distribution of scene information to concentrate on the surface of scenes.

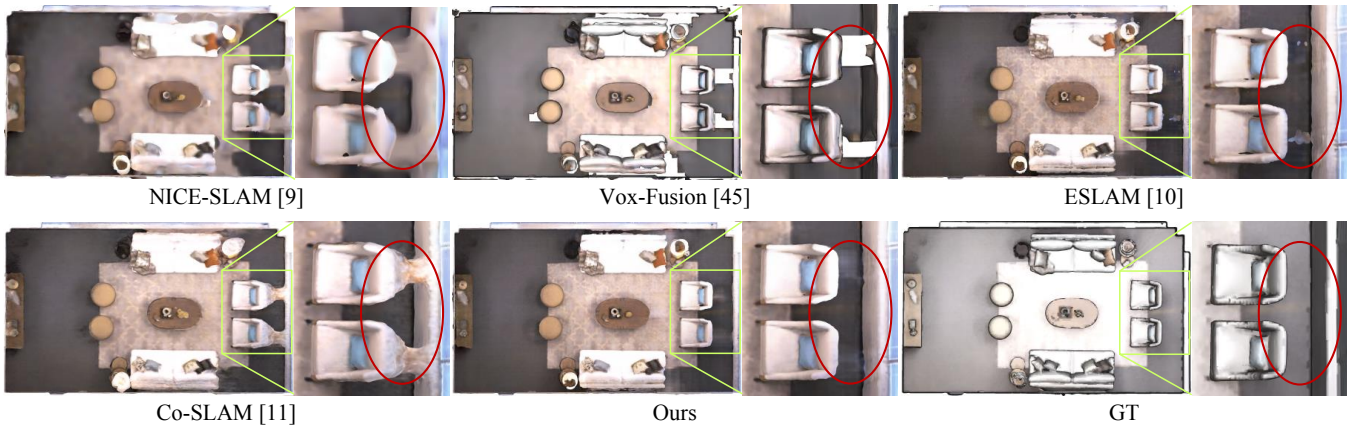


Fig. 3. Reconstruction results of room0. The chair region is magnified to highlight. Compared to other methods, DVN-SLAM can achieve accurate modeling of local details and global structure, providing more reasonable predictions for unobserved regions, such as the backside of the chair.

TABLE I

QUANTITATIVE COMPARISON OF DVN-SLAM WITH EXISTING NERF-BASED DENSE V-SLAM IN TERMS OF RECONSTRUCTION AND LOCALIZATION ACCURACY ON THE REPLICA DATASET [50]. IMAP* DENOTES THE IMAP [9] REIMPLEMENTATION RELEASED BY THE NICE-SLAM [10] AUTHORS.

Methods	Reconstruction				Localization	
	Depth L1[cm] ↓	Acc.[cm] ↓	Comp.[cm] ↓	Comp.Ratio[%] ↑	ATE Mean[cm] ↓	ATE RMSE[cm] ↓
IMAP* [9]	4.64	3.62	4.93	80.51	2.59	3.42
NICE-SLAM [10]	1.90	2.37	2.64	91.13	1.56	2.05
Vox-Fusion [51]	2.91	1.88	2.56	90.93	1.02	1.47
Co-SLAM [12]	1.51	2.10	2.08	93.44	0.94	1.06
ESLAM [11]	<u>0.94</u>	2.18	<u>1.75</u>	96.46	<u>0.52</u>	<u>0.64</u>
DVN-SLAM (Ours)	0.75	<u>2.09</u>	1.70	<u>96.43</u>	0.45	0.53

The total loss function is defined as:

$$L_{total} = \lambda_{rgb}L_{rgb} + \lambda_dL_d + \lambda_{ic}L_{ic} + \lambda_{T-m}L_{T-m} + \lambda_{T-t}L_{T-t} + \lambda_{fs}L_{fs}, \quad (9)$$

where L_{T-m} , L_{T-t} , L_{fs} are TSDF losses from ESLAM [11]. $\{\lambda_{rgb}, \lambda_d, \lambda_{T-m}, \lambda_{T-t}, \lambda_{fs}, \lambda_{ic}\}$ are loss weights.

C. Mapping and Tracking

Given an initial pose and the first frame, we render images from this pose and minimize the total loss to initialize the neural implicit representation.

Tracking. For each observed frame, using the previous frame pose as the source pose, we do rendering and calculate losses, and iteratively optimize to obtain the localization result.

Mapping. Following [11], we apply a keyframe selection every k frames and perform map optimization. We select W frames including the current frame, the previous two keyframes, and $W - 3$ frames randomly selected from the keyframe list. We do rendering and calculate losses, and iteratively optimize the feature planes, fusion network, and corresponding camera poses.

IV. EXPERIMENT

We conduct experiments to demonstrate the superior localization and mapping accuracy, as well as dynamic robustness, of the DVN-SLAM compared to other methods.

A. Experiment Setup

Datasets. Following previous work, we conduct experimental evaluations on two standard 3D datasets: the Replica dataset [50] and the TUM_RGBD dataset [52]. In addition to the static scenes commonly used in previous work, we do experiments in challenging dynamic scenes from the TUM_RGBD dataset [52]. We also conduct experiments on self-collected RGB-D data.

Implementation details. The resolution of coarse and fine feature planes is $\{24 \text{ cm}, 6 \text{ cm}\}$ for geometry and $\{24 \text{ cm}, 3 \text{ cm}\}$ for appearance. The feature dimension is 24. The MLPs in the neural implicit representation consist of two layers with 16 channels, using the ReLU activation function. The truncation distance $tr = 6 \text{ cm}$. The fusion weight $\omega = 0.5$. In the tracking process, we sample 2000 pixels per image. The loss weights are set as $\lambda_{rgb} = 5, \lambda_d = 1, \lambda_{T-m} = 200, \lambda_{T-t} = 50, \lambda_{fs} = 5, \lambda_{ic} = 0$. In the mapping process, we sample 4000 pixels per image. The loss weights are set as $\lambda_{rgb} = 5, \lambda_d = 0.1, \lambda_{T-m} = 200, \lambda_{T-t} = 10, \lambda_{fs} = 5, \lambda_{ic} = 0.05$. All experiments are conducted on a server equipped with an NVIDIA A100 GPU.

Metrics. Following the evaluation methodology of Co-SLAM [12], we evaluate the reconstruction quality using Depth L1 [cm], Accuracy (Acc.[cm]), Completeness (Comp.[cm]), and Completion Rate (Comp. Ratio[%]) thresholds set to 5 cm. For tracking evaluation, we utilize the Absolute Trajectory Error (ATE) mean and Root Mean Square Error (RMSE) in centimeters.

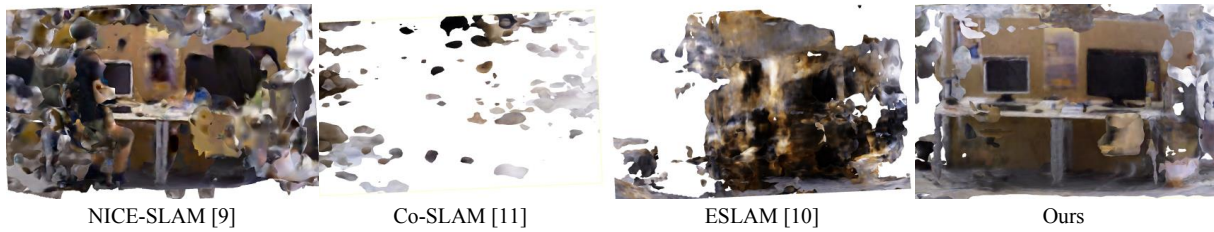


Fig. 4. Reconstruction results on TUM-RGBD. In dynamic scenes, our method effectively ignores dynamic people and successfully reconstructs the static background, while other methods completely fail.

TABLE II

ATE RSME (CM) RESULTS ON TUM-RGBD DATASET. COMPARED TO OTHER NERF-BASED SLAM METHODS, OURS ACHIEVES REMARKABLE LOCALIZATION RESULTS IN STATIC AND DYNAMIC SCENES. “-” FOR POINT-SLAM [44] IN FR3/WALKING-STATIC MEANS A COMPLETE FAILURE.

Methods	fr1/desk	Static fr2/xyz	fr3/office	fr3/walking-halfshere	Dynamic fr3/walking-xyz	fr3/walking-static
iMAP* [9]	7.20	2.10	9.00	9099.5	127.97	1434.9
NICE-SLAM [10]	2.98	1.62	4.05	692.27	120.17	<u>67.94</u>
Co-SLAM [12]	2.98	1.83	2.76	144.86	98.48	88.88
ESLAM [11]	<u>2.54</u>	1.14	2.81	<u>79.24</u>	<u>96.50</u>	70.22
Point-SLAM [44]	4.34	1.31	3.48	124.05	856.39	-
DVN-SLAM (Ours)	2.29	<u>1.25</u>	<u>2.78</u>	51.77	17.68	2.05

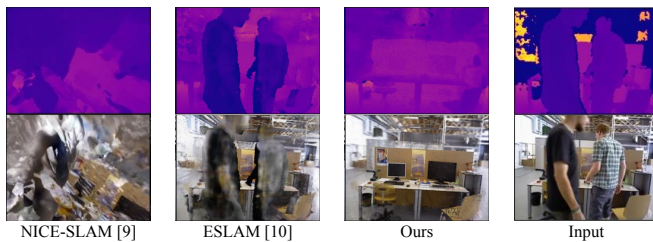


Fig. 5. Rendering results in dynamic scenes. Our method not only produces clearer rendering in static regions but also effectively filters out dynamic objects and fills in the corresponding background.

B. Experiment Results

Evaluation on Replica [50]. We conducted experiments on 8 scenes from the Replica dataset [50]. The average results over 5 runs on 8 scenes are presented in Tab. I. Compared to other Nerf-based SLAM methods, ours achieves state-of-the-art performance in terms of localization accuracy, Depth L1, and completeness, while ranking second in terms of accuracy and completeness ratio. Due to the adoption of an Octree for scene representation in Vox-Fusion [51], the reconstructed results tend to have more holes, resulting in a high accuracy but a low completeness ratio.

Fig. 3 illustrates a visualization comparison of the reconstruction results between DVN-SLAM and others. It can be observed that ours not only captures finer details but also demonstrates superior predictive capabilities for unobserved regions. The chair region is magnified to highlight. In this sequence, the backside of the chair is not observed from any viewpoint. Vox-Fusion [51] utilizes an octree structure for scene representation, resulting in significant holes in the reconstruction. ESLAM [11] employs feature-plane-based scene representation, lacking global prediction capability, leading to holes on the front side of the chair and notice-

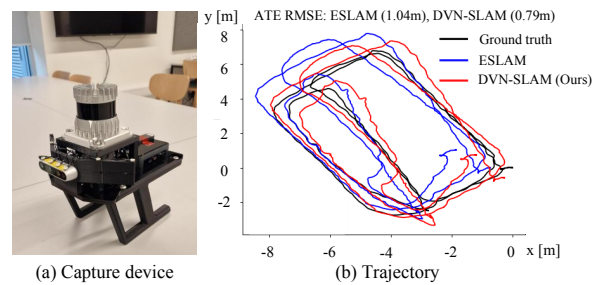


Fig. 6. Capture device and localization result in real-world scenes. Our method proves to be effective when applied in practical scenarios.

able protrusions on the backside. Co-SLAM [12] simply concatenates sparse hash encoding and coordinate encoding, resulting in excessive filling of unobserved areas. In contrast, DVN-SLAM predicts a smooth wall and floor surface.

Evaluation on TUM-RGBD [52]. Tab. II presents the evaluation results in static scenes and dynamic scenes of the TUM-RGBD dataset. Compared to other Nerf-based methods, our approach achieves remarkable performance consistently. Particularly in dynamic scenes, ours remains effective. The poor performance in fr3/walking-halfshere is due to the moving people being too close to the camera, while other methods perform even worse.

Fig. 4 showcases the reconstruction results of the fr3/walking-xyz. Our method successfully ignores fast-moving individuals and reconstructs the static background, while other methods completely fail. Fig. 5 illustrates the rendering visualization during the mapping process. Our method produces clearer rendering in static regions, and it effectively filters out dynamic objects and fills in occluded backgrounds. This is due to the designed local-global fusion neural implicit representation, which makes our method

TABLE III

ABLATION STUDY OF LOCAL-GLOBAL FUSION NEURAL IMPLICIT REPRESENTATION CHOICES ON ROOM0 OF REPLICA [50] AND F3/WALKING-XYZ OF TUM.RGBD [52]. THE RESULTS VALIDATE THE EFFECTIVENESS OF EACH OF OUR INNOVATIONS.

Experiment	room0					fr3/walking-xyz	
	Acc. $\downarrow$ Acc.[cm] $\downarrow$	Comp.[cm] $\downarrow$	Comp.Ratio[%] $\uparrow$	ATE Mean[cm] $\downarrow$	ATE RMSE[cm] $\downarrow$	ATE Mean[cm] $\downarrow$	ATE RMSE[cm] $\downarrow$
a. w/o local representation	90.45	84.31	5.72	110.03	114.74	17.41	24.65
b. w/o global representation	2.38	1.85	97.17	0.57	0.67	19.86	24.93
c. w/o attention-based feature fusion	2.57	2.17	94.93	0.74	0.89	18.09	22.53
d. w/o result fusion	2.37	1.80	97.02	0.49	0.60	13.90	18.05
e. w/ concatenation feature fusion	2.38	1.80	97.10	0.63	0.75	16.85	21.50
f. Full DVN-SLAM	2.32	1.77	97.24	0.45	0.57	13.26	17.68

insensitive to dynamic objects and can maintain the scene structure modeling.

Evaluation on real-world scene. Fig. 6 (a) shows our capture device. The RGB-D camera is the RealSense D435i and the small nodes on top of the camera and the front wings are lasers for the Phasespace motion capture localization system to get the trajectory ground truth. The captured sequence consists of 6677 pairs of RGB-D images with a resolution of 1280×720 and a frequency of 30 Hz. Fig. 6 (b) showcases the localization results. DVN-SLAM exhibits superior accuracy compared to ESLAM [11] in real-world scenes, thereby validating its practicability.

TABLE IV

RUNTIME ANALYSIS OF OUR METHOD IN COMPARISON WITH OTHERS.

Method	iMAP* [9]	NICE-SLAM [10]	Vox-Fusion [51]	Co-SLAM [12]	ESLAM [11]	Ours
Tracking FPS $\uparrow$	9.92	13.70	2.11	17.24	18.11	9.62
Mapping FPS $\uparrow$	2.23	0.20	2.17	10.02	3.62	1.82

C. Runtime analysis.

Our method achieves 7.32 Hz tracking and 1.67 Hz mapping performance on a server with an NVIDIA A100 GPU. Tab. IV illustrates a comparison of the runtime between ours and others. Due to the more complex scene representation, our algorithm exhibits lower real-time performance compared to ESLAM [11] and Co-SLAM [12].

D. Ablation Study

Effect of local-global fusion neural implicit representation. Tab. III presents the quantitative evaluations of different neural implicit representations. Our complete model outperforms scene representation without local-global fusion in terms of reconstruction and localization accuracy. Fig. 7 presents the visualization of the reconstruction results. It can be observed that the complete model exhibits finer reconstructions and more reasonable predictions. The complete failure after removing the local representation is attributed to the difficulty of modeling large scenes using a smaller MLP and a lower number of iterations. Additionally, we employed direct concatenation for feature fusion. As shown in Tab. III, row e, and Fig. III (e), excessive padding is observed on the back of the chair.

Effect of information concentration loss. Fig. 8 illustrates the influence of information concentration loss on reconstruction, showcasing the beneficial optimization of scene information. It can suppress unnecessary reconstructions and

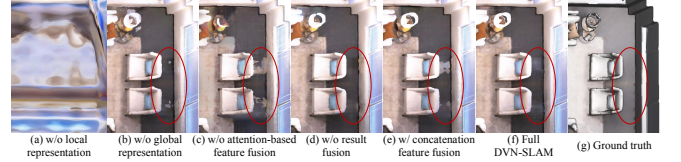


Fig. 7. Visualization of reconstruction in ablation experiments. The back of the chairs is highlighted with red circles, demonstrating the effectiveness of the local-global representation in making reasonable and smooth predictions for unobserved regions.

improve reconstruction accuracy. In addition, we conducted a comparative experiment with deep regularization loss L_{dist} in Mip-NeRF 360 [53]. The motivation behind depth regularization in Mip-NeRF 360 [53] is similar to our information concentration loss. However, ours is obtained by the depth variance along the ray, encouraging information to be more focused on the depth center. It is more concise and the red circle in Fig. 8 shows the better suppression effect of our method for unnecessary reconstruction.

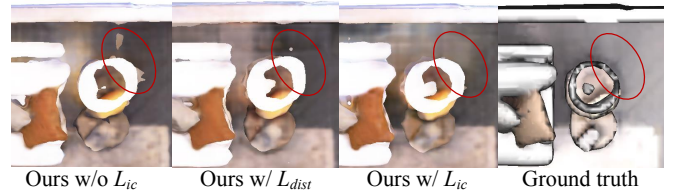


Fig. 8. Visualization of information concentration loss ablation experiment.

V. CONCLUSION

We propose DVN-SLAM, a dynamic robust dense visual SLAM system based on a local-global fusion neural implicit representation. Thanks to the fusion representation and information concentration loss, DVN-SLAM is able to make reasonable predictions for the backside of objects, effectively avoiding holes and overfilling. More importantly, DVN-SLAM is dynamically robust and remains effective in high-dynamic scenes without semantic and optical flow prior. Despite achieving promising results in indoor scenes, our proposed algorithm still faces challenges in larger-scale outdoor scenes. Challenges such as borderlessness, lighting variations, and sensor noise need to be addressed.

REFERENCES

- [1] M. Muglikar, Z. Zhang, and D. Scaramuzza, "Voxel map for visual slam," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2020, pp. 4181–4187.
- [2] A. P. Gee, D. Chekhlov, A. Calway, and W. Mayol-Cuevas, "Discovering higher level structure in visual slam," *IEEE Transactions on Robotics*, vol. 24, no. 5, pp. 980–990, 2008.
- [3] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "Nerf: Representing scenes as neural radiance fields for view synthesis," *Communications of the ACM*, vol. 65, no. 1, pp. 99–106, 2021.
- [4] J. T. Barron, B. Mildenhall, M. Tancik, P. Hedman, R. Martin-Brualla, and P. P. Srinivasan, "Mip-nerf: A multiscale representation for anti-aliasing neural radiance fields," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 5855–5864.
- [5] A. Pumarola, E. Corona, G. Pons-Moll, and F. Moreno-Noguer, "D-nerf: Neural radiance fields for dynamic scenes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 10 318–10 327.
- [6] R. Martin-Brualla, N. Radwan, M. S. Sajjadi, J. T. Barron, A. Dosovitskiy, and D. Duckworth, "Nerf in the wild: Neural radiance fields for unconstrained photo collections," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 7210–7219.
- [7] Z. Yu, S. Peng, M. Niemeyer, T. Sattler, and A. Geiger, "Monosdf: Exploring monocular geometric cues for neural implicit surface reconstruction," *Advances in neural information processing systems*, vol. 35, pp. 25 018–25 032, 2022.
- [8] G. Wang, L. Pan, S. Peng, S. Liu, C. Xu, Y. Miao, W. Zhan, M. Tomizuka, M. Pollefeys, and H. Wang, "Nerf in robotics: A survey," *arXiv preprint arXiv:2405.01333*, 2024.
- [9] E. Sucar, S. Liu, J. Ortiz, and A. J. Davison, "imap: Implicit mapping and positioning in real-time," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 6229–6238.
- [10] Z. Zhu, S. Peng, V. Larsson, W. Xu, H. Bao, Z. Cui, M. R. Oswald, and M. Pollefeys, "Nice-slam: Neural implicit scalable encoding for slam," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 12 786–12 796.
- [11] M. M. Johari, C. Carta, and F. Fleuret, "Eslam: Efficient dense slam system based on hybrid representation of signed distance fields," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 17 408–17 419.
- [12] H. Wang, J. Wang, and L. Agapito, "Co-slam: Joint coordinate and sparse parametric encodings for neural real-time slam," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13 293–13 302.
- [13] C. Ruan, Q. Zang, K. Zhang, and K. Huang, "Dn-slam: A visual slam with orb features and nerf mapping in dynamic environments," *IEEE Sensors Journal*, 2023.
- [14] Z. Xu, J. Niu, Q. Li, T. Ren, and C. Chen, "Nid-slam: Neural implicit representation-based rgb-d slam in dynamic environments," *arXiv preprint arXiv:2401.01189*, 2024.
- [15] N. Schischka, H. Schieber, M. Asim Karaoglu, M. Görgülü, F. Grötzner, A. Ladikos, D. Roth, N. Navab, and B. Busam, "Dynamon: Motion-aware fast and robust camera localization for dynamic neural radiance fields," *arXiv e-prints*, pp. arXiv–2309, 2023.
- [16] M. Li, J. He, G. Jiang, and H. Wang, "Ddn-slam: Real-time dense dynamic neural implicit slam with joint semantic encoding," *arXiv preprint arXiv:2401.01545*, 2024.
- [17] G. Wang, L. Song, Y. Shen, and H. Wang, "3d multi-object tracking based on simultaneous optimization of object detection and scene flow estimation," *ROBOT*, vol. 46, no. 5, pp. 554–561, 2024.
- [18] R. A. Newcombe, S. J. Lovegrove, and A. J. Davison, "Dtam: Dense tracking and mapping in real-time," in *2011 international conference on computer vision*. IEEE, 2011, pp. 2320–2327.
- [19] R. A. Newcombe, S. Izadi, O. Hilliges, D. Molyneaux, D. Kim, A. J. Davison, P. Kohi, J. Shotton, S. Hodges, and A. Fitzgibbon, "Kinectfusion: Real-time dense surface mapping and tracking," in *2011 10th IEEE international symposium on mixed and augmented reality*. Ieee, 2011, pp. 127–136.
- [20] P. J. Besl and N. D. McKay, "Method for registration of 3-d shapes," in *Sensor fusion IV: control paradigms and data structures*, vol. 1611. Spie, 1992, pp. 586–606.
- [21] T. Whelan, S. Leutenegger, R. Salas-Moreno, B. Glocker, and A. Davison, "Elasticfusion: Dense slam without a pose graph." Robotics: Science and Systems, 2015.
- [22] T. Schöps, T. Sattler, and M. Pollefeys, "Surfelmeshing: Online surfel-based mesh reconstruction," *IEEE transactions on pattern analysis and machine intelligence*, vol. 42, no. 10, pp. 2494–2507, 2019.
- [23] E. Vespa, N. Nikolov, M. Grimm, L. Nardi, P. H. Kelly, and S. Leutenegger, "Efficient octree-based volumetric slam supporting signed-distance and occupancy mapping," *IEEE Robotics and Automation Letters*, vol. 3, no. 2, pp. 1144–1151, 2018.
- [24] B. Xu, W. Li, D. Tzoumanikas, M. Bloesch, A. Davison, and S. Leutenegger, "Mid-fusion: Octree-based object-level multi-instance dynamic slam," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 2019, pp. 5231–5237.
- [25] T. Schops, T. Sattler, and M. Pollefeys, "Bad slam: Bundle adjusted direct rgb-d slam," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 134–144.
- [26] M. Bloesch, J. Czarnowski, R. Clark, S. Leutenegger, and A. J. Davison, "Codeslam—learning a compact, optimisable representation for dense visual slam," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 2560–2568.
- [27] R. Li, S. Wang, and D. Gu, "Deepslam: A robust monocular slam system with unsupervised deep learning," *IEEE Transactions on Industrial Electronics*, vol. 68, no. 4, pp. 3577–3587, 2020.
- [28] L. Koestler, N. Yang, N. Zeller, and D. Cremers, "Tandem: Tracking and dense mapping in real-time using deep multi-view stereo," in *Conference on Robot Learning*. PMLR, 2022, pp. 34–45.
- [29] S. Peng, M. Niemeyer, L. Mescheder, M. Pollefeys, and A. Geiger, "Convolutional occupancy networks," in *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part III 16*. Springer, 2020, pp. 523–540.
- [30] Z. Teed and J. Deng, "Droid-slam: Deep visual slam for monocular, stereo, and rgb-d cameras," *Advances in neural information processing systems*, vol. 34, pp. 16 558–16 569, 2021.
- [31] W. Wu, G. Wang, J. Zhong, H. Wang, and Z. Liu, "Self-supervised multi-frame monocular depth estimation with pseudo-lidar pose enhancement," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 10 018–10 025.
- [32] G. Wang, J. Zhong, S. Zhao, W. Wu, Z. Liu, and H. Wang, "3d hierarchical refinement and augmentation for unsupervised learning of depth and pose from monocular video," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 4, pp. 1776–1786, 2022.
- [33] M. Tancik, V. Casser, X. Yan, S. Pradhan, B. Mildenhall, P. P. Srinivasan, J. T. Barron, and H. Kretschmar, "Block-nerf: Scalable large scene neural view synthesis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 8248–8258.
- [34] H. Turki, D. Ramanan, and M. Satyanarayanan, "Mega-nerf: Scalable construction of large-scale nerfs for virtual fly-throughs," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 12 922–12 931.
- [35] H. Turki, J. Y. Zhang, F. Ferroni, and D. Ramanan, "Suds: Scalable urban dynamic scenes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 12 375–12 385.
- [36] Z. Jia, B. Wang, and C. Chen, "Drone-nerf: Efficient nerf based 3d scene reconstruction for large-scale drone survey," *Image and Vision Computing*, vol. 143, p. 104920, 2024.
- [37] W. Wu, Q. Wang, G. Wang, J. Wang, T. Zhao, Y. Liu, D. Gao, Z. Liu, and H. Wang, "Emie-map: Large-scale road surface reconstruction based on explicit mesh and implicit encoding," in *European Conference on Computer Vision*. Springer, 2024, pp. 370–386.
- [38] Z. Li, S. Niklaus, N. Snively, and O. Wang, "Neural scene flow fields for space-time view synthesis of dynamic scenes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 6498–6508.
- [39] Z. Yan, C. Li, and G. H. Lee, "Nerf-ds: Neural radiance fields for dynamic specular objects," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 8285–8295.
- [40] X. Chen, Q. Zhang, X. Li, Y. Chen, Y. Feng, X. Wang, and J. Wang, "Hallucinated neural radiance fields in the wild," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 12 943–12 952.

- [41] T. Müller, A. Evans, C. Schied, and A. Keller, "Instant neural graphics primitives with a multiresolution hash encoding," *ACM Transactions on Graphics (ToG)*, vol. 41, no. 4, pp. 1–15, 2022.
- [42] F. Tosi, Y. Zhang, Z. Gong, E. Sandström, S. Mattoccia, M. R. Oswald, and M. Poggi, "How nerfs and 3d gaussian splatting are reshaping slam: a survey," *arXiv preprint arXiv:2402.13255*, vol. 4, 2024.
- [43] T. Deng, G. Shen, T. Qin, J. Wang, W. Zhao, J. Wang, D. Wang, and W. Chen, "Plgslam: Progressive neural scene representation with local to global bundle adjustment," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 19 657–19 666.
- [44] E. Sandström, Y. Li, L. Van Gool, and M. R. Oswald, "Point-slam: Dense neural point cloud-based slam," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 18 433–18 444.
- [45] C. Yan, D. Qu, D. Xu, B. Zhao, Z. Wang, D. Wang, and X. Li, "Gs-slam: Dense visual slam with 3d gaussian splatting," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 19 595–19 604.
- [46] H. Matsuki, R. Murai, P. H. Kelly, and A. J. Davison, "Gaussian splatting slam," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 18 039–18 048.
- [47] T. Deng, Y. Chen, L. Zhang, J. Yang, S. Yuan, D. Wang, and W. Chen, "Compact 3d gaussian splatting for dense visual slam," *arXiv preprint arXiv:2403.11247*, 2024.
- [48] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3d gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, pp. 139–1, 2023.
- [49] T. Müller, B. McWilliams, F. Rousselle, M. Gross, and J. Novák, "Neural importance sampling," *ACM Transactions on Graphics (ToG)*, vol. 38, no. 5, pp. 1–19, 2019.
- [50] J. Straub, T. Whelan, L. Ma, Y. Chen, E. Wijmans, S. Green, J. J. Engel, R. Mur-Artal, C. Ren, S. Verma, *et al.*, "The replica dataset: A digital replica of indoor spaces," *arXiv preprint arXiv:1906.05797*, 2019.
- [51] X. Yang, H. Li, H. Zhai, Y. Ming, Y. Liu, and G. Zhang, "Vox-fusion: Dense tracking and mapping with voxel-based neural implicit representation," in *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 2022, pp. 499–507.
- [52] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A benchmark for the evaluation of rgb-d slam systems," in *2012 IEEE/RSJ international conference on intelligent robots and systems*. IEEE, 2012, pp. 573–580.
- [53] J. T. Barron, B. Mildenhall, D. Verbin, P. P. Srinivasan, and P. Hedman, "Mip-nerf 360: Unbounded anti-aliased neural radiance fields," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5470–5479.